MIL-T-12839D(ME)
17 July 1981
SUPERSEDING
MIL-T-12839C(ME)
25 May 1976

THERMOMETERS, SELF-INDICATING, LIQUID-IN-GLASS BASE-LINE TAPE; WITH CASE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers etched-stem, liquid-in-glass thermometers housed in metal cases with clips for attaching to a measuring tape.

1.2 Classification. Thermometers shall be the following types, as specified (see 6.2):

Type I - Graduated in Fahrenheit (F) degrees.
Type II - Graduated in Celsius (C) degrees.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6685

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The thermometers shall be etched-stem, liquid-in-glass, total-immersion type, with a protective case, as shown in Figure 1, and as specified herein.

3.2 Material. Material shall be as specified herein and as shown on Figure 1. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

3.3 Storage conditions. The thermometers shall not be damaged by storage in any ambient temperature from minus 30° F to plus 160° F (minus 34° C to plus 71° C).

3.4 Thermometer tube.

3.4.1 Stem. The stem shall be magnifying glass type, and shall have a broad strip of white glass at the back to increase reflected light. The stem

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shall be designed to increase the angle of vision and effective width of the thermosensitive column and to accommodate the graduations and numbers on the flat front of the tube. The stem shall be annealed.

3.4.2 Thermosensitive fluid. The bulb and tube shall contain a sufficient quantity of fluid so that, when correctly calibrated, the height of the fluid in the tube shall indicate the temperature of the medium surrounding the thermometer within the range and accuracy specified herein. The type of fluid used shall be such as to minimize any tendency for the liquid column to separate. A non-fading dye may be used to color the fluid to allow a contrast for reading.

3.4.3 Graduations. The graduation marks shall be etched into the stem and shall be clear, uniform, and symmetrical. Graduation marks and figures shall be filled with a black indelible pigment and shall be clearly legible from a distance of 18 inches. The pigment shall not be affected by total immersion for 10 days in a solution of not less than 75 percent sulfuric acid or a commercial grade of toluene at plus 100° F (plus 38° C).

3.5 Identification marking. Each thermometer and its case shall be identified in accordance with MIL-STD-130.

3.6 Type I - graduated in Fahrenheit degrees. The Fahrenheit scale shall conform to the following characteristics:

(a)	Temperature range	- minus 20° F to plus 120° F.
(b)	Graduations	- 1° F.
(c)	Longer graduation lines at each	- 5° F.
(d)	Graduations numbered at each multiple of	- 10° F.
(e)	Standardization	- minus 10° F, plus 32° F, plus 110° F.
(f)	After standardization, scale error at any point not more than	- 1° F.
(g)	Thermometer inscription	- letter "F" above upper limit of scale graduations.

3.7 Type II - graduated in Celsius degrees. The Celsius scale shall conform to the following characteristics:

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(a)	Temperature range	- minus 30° C to plus 50° C.
(b)	Graduations	- 0.5° C.
(c)	Longer graduation lines at each	- 5° C.
(d)	Graduations numbered at each multiple of	- 5° C.
(e)	Standardization	- minus 10° C, 0° C, plus 45° C.
(f)	After standardization, scale error at any point not more than	- 0.5° C.
(g)	Thermometer inscription	- letter "C" above upper limit of scale graduations.

3.8 Workmanship. The thermometers and thermosensitive fluid shall be free from gas bubbles and foreign material. The thermometers shall be clean and free from pits, chips, sharp edges or other defects which would prove detrimental to accuracy and appearance. The cases shall be free from extraneous material and all edges shall be rounded or beveled. Soldering shall be free from cracks, lack of fusion with parent metal, lack of prescribed fit, or spatter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

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4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 2.5 defective for major defects and 4.0 percent defective for the minor defect.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The thermometers shall be examined as specified herein for the following defects:

Major

101. Material not as specified.
102. Stem not as specified.
103. Temperature range not as specified.
104. Graduations not as specified.
105. Separation or bubbles in the thermosensitive fluid.
106. Case not as specified.
107. Workmanship not as specified.

Minor

201. Identification marking missing, incomplete, or illegible.

4.4.2 Tests.

4.4.2.1 Storage. Sample thermometers shall be placed in a cold chamber at a temperature of minus 30° F (minus 34° C) for not less than 72 hours without solar radiation. The thermometers shall then be removed, allowed to return to room temperature, and placed in an oven at plus 160° F (plus 70° C) for 4 hours. The thermometers shall be removed from the oven, allowed to return to room temperature, and examined. Any evidence of breaks, cracks, or separation in the thermosensitive fluid shall constitute failure of this test.

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4.4.2.2 Marking. Each of the sample thermometers shall be submerged in a solution of not less than 75 percent sulfuric acid or a commercial grade of toluene at plus 100° F (plus 38° C) for a period of 10 days. At the end of the test period, the sample thermometer shall be examined. Any damage or diminution of the marking shall constitute failure of this test.

4.4.2.3 Standardization.

4.4.2.3.1 Type I - graduated in Fahrenheit degrees. Using a thermometer which has been certified by the National Bureau of Standards to have an accuracy of plus or minus 0.1° F, totally immerse the certified thermometer and the sample thermometers in a well-stirred water bath and compare the certified and sample thermometers' readings at minus 10° F, plus 32° F, and plus 110° F. Failure of the sample thermometers to indicate the same reading as the certified thermometer within 1° F shall constitute failure of this test.

4.4.2.3.2 Type II - graduated in Celsius degrees. Using a thermometer which has been certified by the National Bureau of Standards to have an accuracy of plus or minus 0.1° C, totally immerse the certified thermometer and the sample thermometers in a well-stirred water bath, and compare the certified and sample thermometers' readings at minus 10° C, 0° C, and plus 45° C. Failure of the sample thermometers to indicate the same reading as the certified thermometer within 0.5° C shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108.	Methods and containers not as specified.	Each incorrect method or container shall constitute one defect.
109.	Quantity placed in consolidated container not as specified for Level A or Level B.	
110.	Thermometers of unlike type consolidated together for Level A.	

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111. Fiberboard boxes not closed as specified for Level A or Level B.
112. Commercial packaging not in accordance with referenced documents.
113. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each thermometer, in its case, shall be preserved in accordance with MIL-P-116, Method III.

5.1.1.1 Consolidation. Thermometers, of like type, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, style optional, in quantities as specified (see 6.2). Box closure shall be in accordance with the appendix to the box specification.

5.1.2 Commercial. Thermometers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. The thermometers preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, Style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. The thermometers preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-636, V3c, style optional. The boxes shall be closed in accordance with Method V of the appendix to the box specification.

5.2.3 Commercial. Thermometers shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for Military levels of protection (Level A or Level B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The thermometers are intended for use with precise surveying tapes in measuring survey base lines.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of thermometer required (see 1.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) Quantities to be consolidated together (see 5.1.1.1).

6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

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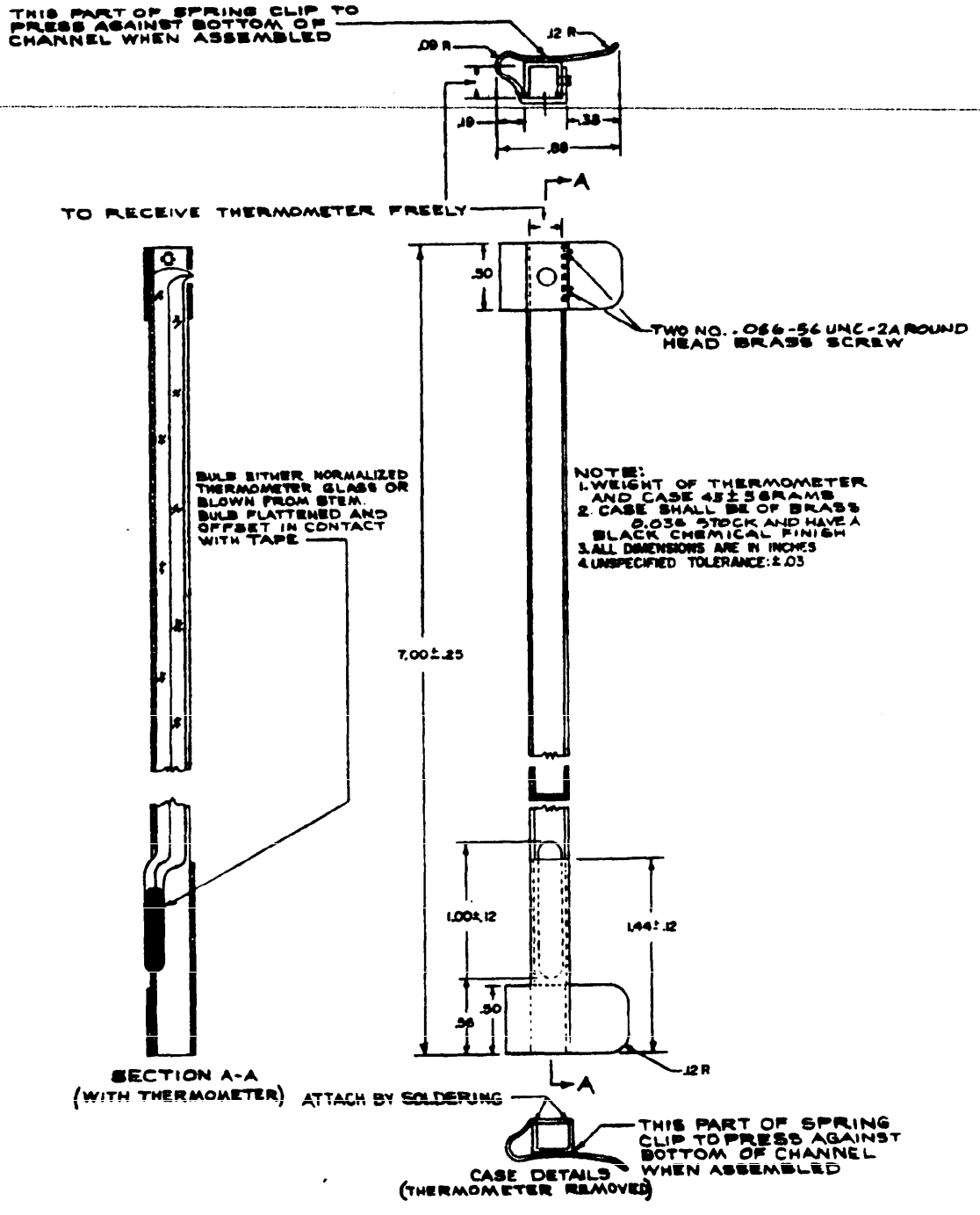


FIGURE 1. Base line thermometer.

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